MILITARY SPECIFICATION
SWITCH, HYDROSTATIC, MARK 41 MODS 0 AND 1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of the Switch, Hydrostatic, Mark 41 Mods 0 and 1, Naval Sea Systems Command LD 299449 and PL 3212233, respectively.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code ident 10001)

LD 299449	Switch, Hydrostatic, Mark 41 Mod 0
PL 3212233	Switch, Hydrostatic, Mark 41 Mod 1
DL 2644484	Shipping (Packing) Condition Hydrostatic Switch

All drawing numbers referenced herein are listed on LD 299449 or PL 3212233 unless otherwise designated.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

WR-43	Naval Weapons Requirements, Preparation of Quality Assurance Requirements

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

HANDBOOK H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3 REQUIREMENTS

3.1 Description. The Switch, Hydrostatic, Mark 41 Mods 0 and 1 is a normally-open, four-circuit, make and break device including a multiple terminal connector plug whereby the switch

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is connected with external circuitry. The switch is locked in the safe or open position by means of a safety pin. Upon removal of this pin, a diaphragm and piston assembly of the switch is freed to close the switch as a predetermined hydrostatic pressure is applied to the sea side of the device. A spring beneath the piston and diaphragm serves to return the switch to the open position as the external pressure is released.

3.2 First article sample. Unless otherwise specified in the contract or order, a first article sample of the switches shall be manufactured and prepared for delivery using the methods and procedures proposed for the production lot. The sample is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing items that comply with this specification and the design. The sample size shall be as specified in the contract or purchase order.

3.3 General requirements.

3.3.1 Fabrication. The switch shall be manufactured in accordance with LD 299449 or PL 3212233, as applicable, and as specified herein.

3.3.2 Interchangeability. Components parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in accordance with applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection for fit or performance.

3.3.3 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.3.4 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

3.4 Performance requirements and product characteristics. The switch shall meet the following performance requirements and product characteristics:

3.4.1 Bridge wire. After soldering the bridge wire between the two center terminals (I and J) of the connector, the bridge wire shall withstand a pull of 8.9 newtons (N) (2-pound force) applied outwardly along the axis of the connector without loosening or affecting the electrical resistance between the connected terminals. After an 8.9-N (2-pound) pull on the bridge wire, the resistance between Terminals I and J shall not exceed 0.1 ohm. See 6.4.

3.4.2 Cap assembly. Before and after switch assembly, the cap shall be capable of 365 degrees manual rotation within the bell of the cap assembly without binding.

3.4.3 Piston flange. At the time of assembly of the diaphragm with the piston, the edge of the undercut inner groove in the piston flange, by means of which the inner edge of the diaphragm is retained, shall be free of a sharp edge and showing the radius as called for on the drawings.

3.4.4 Retaining rings. Following switch assembly, all retaining rings shall be fully seated in their respective grooves. To insure proper seating of the rings, the diaphragm shall, in each instance, be compressed sufficiently to clear the ring groove before the rings are mounted.

3.4.5 Safety pin. At atmospheric pressure, the safety pin when inserted through the holes provided in the cap shall engage the groove on the piston in a manner to lock the piston in its retracted (open-circuit) position.

3.4.6 Torque. Following switch assembly, the Nuts, Drawing 2418050, shall withstand without movement a torque of 0.452N-metre (N·m) (4.0 pound-inches) maximum applied in the same direction in which the nuts were tightened.

3.4.7 Leakage.

3.4.7.1 Mark 41 Mod 0. The switch shall withstand without leakage air pressures of 83 and 407 kilopascals (kPa) (12 and 59 PSIG).

3.4.7.2 Mark 41 Mod 1. The switch shall withstand without leakage air pressures of 55 and 407 kPa (8.0 and 59 PSIG).

3.4.8 Pressure response.

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3.4.8.1 Mark 41 Mod 0. All four switch circuits shall remain open at all pressures below 211 kPa (30.6 PSIG) and shall close when a pressure of 211 to 261 kPa (30.6 to 37.8 PSIG) is applied to the sea side of the diaphragm.

3.4.8.2 Mark 41 Mod 1. All four switch circuits shall remain open at all pressures below 34.5 kPa (5.0 PSIG) and shall close when a pressure of 34.5 to 62.1 kPa (5.0 to 9.0 PSIG) is applied to the sea side of the diaphragm.

3.4.9 Insulation resistance. The insulation resistance between each of the conductive elements and between each of the conductive elements and the remainder of the metallic elements of the switch shall be 50 megohms or greater when measured at a potential at 500 ± 50 volts DC.

3.4.10 Contact resistance. The contact resistance of each switch circuit, with the switch closed, shall be no greater than 0.2 ohm as measured under a maximum potential of 5.5 volts DC.

3.4.11 Operating position. Unless otherwise specified, the switch shall operate in a manner to meet all requirements of this specification in the following positions: (a) sea side up; (b) sea side down; (c) switch axis horizontal.

3.5 Environmental requirements.

3.5.1 Vibration. The switch shall meet all performance requirements and product characteristics of this specification after being vibrated in accordance with 4.8.1.

3.5.2 Shock. The hydrostatic switch shall meet all performance requirements and product characteristics of this specification after being shocked in accordance with 4.8.2.

3.5.3 Temperature.

3.5.3.1 Operating temperature. The hydrostatic switch shall operate in a manner to meet all performance requirements and product characteristics of this specification at any temperature between -3°C and 34°C (26°F and 93°F).

3.5.3.2 Storage temperature. When packaged and packed in accordance with section 5, the hydrostatic switch shall withstand without damage one 14-day JAN TEMPERATURE AND HUMIDITY CYCLE defined in MIL-STD-331, Test Method 105, except that the relative humidity in the chamber need not exceed room ambient or ambient as controlled.

3.5.4 High pressure.

3.5.4.1 Ruggedness. The switch shall withstand without damage 4.14 MPa (600 PSIG) hydrostatic pressure for three minutes and show no evidence of leakage.

3.5.4.2 Safety. When the switch, with the safety pin in place, is subjected to a hydrostatic pressure of 4.14 MPa (600 PSIG) for three minutes, the switch shall not become unsafe by closing of the switches.

3.6 Cleanup. Prior to final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter.

3.7 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the hydrostatic switch shall be classified as follows:

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a. First Article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
c. Periodic production sample inspection (see 4.6)

4.3 Test equipment. The following items of test equipment are required to perform the tests required in this specification:

a. Test Set Mark 250 type with Accessory Set Mark 5 Mod 0 and Accessory Set Mark 13 Mod 0 (Government furnished) (see 6.2d).

b. A torque wrench capable of applying .565 N•m (5 pound-inches) of torque with an accuracy of ± 2 percent.

c. A test chamber capable of withstanding an internal pressure of 4.14 MPa (600 PSIG) hydrostatic pressure and capable of accepting a switch.

d. A pull tester capable of measuring a pull of 8.9-N to 22.2-N (2 to 5 pounds).

e. A resistance bridge capable of measuring a resistance of 1 ohm in increments of 0.1 ohm, accuracy 1 percent.

4.4 First article inspection.

4.4.1 First article sample. First articles shall be subjected to the examinations and tests designated in Table I. First article sample approval will be granted when the Government testing activity, through testing and subsequent analysis of the test results, is assured of contractor conformance to the contract technical requirements. Conditional acceptance may be granted at the option of the Government based on analysis of detected defects. A first article sample may be disapproved when either nonconforming characteristics or unauthorized materials are found. (See 6.2b.)

4.5 Quality conformance inspection.

4.5.1 100% inspection. During assembly, each unit shall be subjected to the examinations and tests designated in Table I. Any unit containing one or more defects shall be cause for regarding the unit as defective.

4.5.2 Sampling. Tests designated as sampling shall be conducted on a random sample of switches selected from each production lot (see 6.3.2) in accordance with MIL-STD-105, Inspection Level II, AQL of 1.0 percent defective. Unless otherwise specified (see 6.2f), a production lot shall consist of one month's production. Lot composition shall be in accordance with MIL-STD-105, Section 5. Failure of any unit to pass any quality conformance sampling test shall reject the defective unit and shall result in rejection of the lot it represents if the number of defective units equals or exceeds the rejection number for the stated AQL.

4.6 Periodic production inspection. (see 6.2g).

4.6.1 Periodic production samples. Periodic production tests specified in Table I shall be conducted on a random sample of hydrostatic switches selected from each lot (see 6.2f) of hydrostatic switches offered for acceptance. The sample size shall be as specified in the contract or purchase order (see 6.2i). When a unit fails to pass any periodic production test, such failure shall be cause for rejection of the periodic production sample and the lot it represents.

4.7 Examination and test methods

4.7.1 Preparation for delivery. The packaging shall be examined to determine conformance to 5 herein.

4.7.2 Visual examination. Each switch shall be visually examined in accordance with Drawing 1475914 or Drawing 3212233 to determine compliance with the specification and applicable drawings.

4.7.3 Soldering examination (M101). (See 4.9). Examine all soldered connections for excess solder or cold solder joints. Look for loose solder between the terminal pins. Observe for extended discoloration of the contacts as caused from overheating.

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TABLE I - INSPECTION PROCEDURES (1)

Title of Examination or Test	Requirement Paragraph	Method Paragraph	INSPECTION CLASSIFICATIONS		
			First Article	Quality Conformance	Periodic Production
Preparation for delivery	5	4.7.1	1		1
Visual	3.3.1, 3.3.2, 3.6, and 3.7	4.7.2	2 (2)	(3)	2 (2)
Cap assembly	3.4.2	4.7.5	3	(3)	3
Safety pin	3.4.5	4.7.8	4	(3)	4
Leakage	3.4.7	4.7.10 (6)	5	(3)	5
Pressure response	3.4.8	4.7.11.1 (6)	6	(3)	6
Operating position	3.4.11	4.7.11	7	(3)	7
Insulation resistance	3.4.9	4.7.11.2 (6)	8	(3)	8
Contact resistance	3.4.10	4.7.11.3 (6)	9	(3)	9
Torque	3.4.6	4.7.9 (7)	10	(3)	10
Environmental (3 groups)		4.8			
Group (A)					
Vibration	3.5.1	4.8.1	11 (4)		11 (4)
Shock	3.5.2	4.8.2	12 (4)		12 (4)
Group (B)					
Storage Temperature	3.5.3.2	4.8.3.2	13 (4)		13 (4)
Group (C)					
Operating Temperature (High and Low)	3.5.3.1	4.8.3.1	14 (4)		14 (4)
Ruggedness	3.5.4.1	4.8.4.1	15 (5)		15 (5)
Safety	3.5.4.2	4.8.4.2	16		16
Soldering Examination	3.3.4	4.7.3	17	(3)	17
Bridge wire	3.4.1	4.7.4	18	(3)	18
Piston flange	3.4.3	4.7.6	19	(3)	19
Retaining rings	3.4.4	4.7.7	20	(3)	20
Threads	3.3.3	4.7.2	21	(3)	21

NOTES

(1) Inspections shall be performed in successive numerical order for the first and periodic production samples.

(2) These inspections shall be performed as applicable without disassembly of the switch.

(3) Quality conformance inspection shall be an in-process inspection performed during assembly of the switches prior to delivery. The sequence of inspection shall be at the discretion of the manufacturer.

(4) Repeat tests 5 through 9 after vibration, shock, and group B tests.

(5) Repeat tests 5, 8, and 9 after ruggedness tests.

(6) The test method used shall be the one designated herein for the "mod" of the switch being tested.

(7) This is a sampling test.

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4.7.4 Bridge wire test (M102). After the soldering operation securing the bridge wire within the contact assembly, apply an 8.9 + 0, - 0.9N (2 + 0, - 0.2 pounds) pull to the bridge wire in a direction along the axis of the connector assembly. After the pull test verify that the bridge wire is not loose. Using the Resistance Bridge, Item 4.3e., verify that the resistance between terminals I and J does not exceed 0.1 ohm.

4.7.5 Cap Assembly test (M103). With the Retaining Ring, MS 16624-5056, securing the cap within the bell, hold the bell and manually rotate the cap at least 365 degrees within the bell while verifying that there is no binding between the cap and bell.

4.7.6 Piston flange test (M104). Before assembly of the diaphragm with the piston, examine carefully the edge of the inner groove provided in the piston flange, and against which the diaphragm is clamped, for the presence of a sharp cutting edge. Verify that the radius of the groove is in accordance with the applicable drawing.

4.7.7 Retaining ring test (M105). After assembling the diaphragm to the piston, examine the retaining ring to assure that it is properly seated within the piston groove. After final assembly of the switch, inspect the retaining ring securing the cap assembly and diaphragm to the body to assure that it is properly seated within the body groove.

4.7.8 Safety pin test (M106). At atmospheric pressure, observe that the safety pin engages the notch in the piston when the safety pin is inserted and that the piston is locked in the retracted (open-circuit) position.

4.7.9 Torque test (Sampling) (M107). Using the Torque Wrench, Item 4.3b., apply in a tightening direction a torque of 0.452 N·m (4 pound-inches) to the nuts securing the contact assembly to the body. Observe the nuts for turning during the torque application.

4.7.10 Leakage test.

4.7.10.1 Mark 41 Mod 0 (M108). Mount the switch, with the safety pin removed, in the Test Set Mark 250 with sea side of the switch down and exposed to the pressure of the pressure chamber. Cover the other side of the switch with the plastic chamber connected to the manometer. Keep the secondary opening in the plastic chamber open when the pressure in the pressure chamber is changing and closed during the testing interval when the pressure in the pressure chamber is constant. Increase the pressure in the pressure chamber to 83 ± 3.5 kPa (12 ± 0.5 PSIG). Allow 30 seconds for creep of the diaphragm; then close the secondary opening in the plastic chamber. Movement of the oil in the sloping section of the manometer tube at a rate of 25mm (1.0 inch) or more in 30 seconds indicates leakage. With the device so mounted, increase the pressure in the pressure chamber to 407 kPa (59 PSIG). Movement of the oil in the sloping section of the manometer at a rate of 25mm (1.0 inch) or more in 30 seconds indicates leakage. During the foregoing test maintain the switch and the plastic cover at a constant temperature to prevent erroneous results in the manometer reading. Remove the plastic chamber and leave the switch in the test chamber.

4.7.10.2 Mark 41 Mod 1 (M109). Mount the switch, with safety pin removed, in the Test Set Mark 250 with sea side of the switch down and exposed to the pressure of the pressure chamber. Cover the other side of the switch with the plastic chamber connected to the manometer. Keep the secondary opening in the plastic chamber open when the pressure in the pressure chamber is changing and closed during the testing interval when the pressure in the pressure chamber is constant. Increase the pressure in the pressure chamber to 55 kPa (8.0 PSIG). Allow 30 seconds for creep of the diaphragm; then close the secondary opening in the plastic chamber. Movement of the oil in the sloping section of the manometer tube at a rate of 25mm (1.0 inch) or more in 30 seconds indicates leakage. With the device so mounted, increase the pressure in the pressure chamber to 407 kPa (59 PSIG). Movement of the oil in the sloping section of the manometer at a rate of 25mm (1.0 inch) or more in 30 seconds indicates leakage. During the foregoing test maintain the switch and the plastic cover at a constant temperature to prevent erroneous results in the manometer reading. Remove the plastic chamber and leave the switch in the test chamber.

4.7.11 Operating position test (M110. Divide each lot or sample into three equal groups. Subject one group to the following tests in the sea-side-up position, one group in the sea-side-down position, and one group with the longitudinal axis in the horizontal position.

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4.7.11.1 Pressure response tests.

4.7.11.1.1 Mark 41 Mod 0 (M111). With the switch mounted in the Test Set Mark 250, turn the PRESSURE CONTROL to SLOW INTAKE and increase the pressure in the pressure pot to 211 kPa (30.6 PSIG) and then to 261 kPa (37.8 PSIG). Verify the closure of all four switches in the hydrostatic switch at the higher pressure. Reduce the pressure in the pressure pot to zero PSIG.

4.7.11.1.2 Mark 41 Mod 1 (M112). With the switch mounted in the Test Set Mark 250, turn the PRESSURE CONTROL to SLOW INTAKE and increase the pressure in the pressure pot to 34.5 kPa (5.0 PSIG) and then to 62 kPa (9.0 PSIG). Verify the closure of all four switches in the hydrostatic switch at the higher pressure. Reduce the pressure in the pressure pot to zero PSIG.

4.7.11.2 Insulation resistance test.

4.7.11.2.1 Mark 41 Mod 0 (M113). Place each assembled switch within a conditioning chamber for a period of 24 hours while maintaining the chamber at a temperature of 25° ± 1°C (77°F ± 2°F) and a relative humidity of 75 ± 5 percent. Remove the switch from the chamber and place the switch in the pressure pot of the Test Set Mark 250 in the same manner as in 4.7.10. The pressure in the pot shall remain at atmospheric during the first part of this test. Plug the Cable Assembly CA-963 into the receptacle of the switch and rotate the TEST SELECTOR to the INSULATION RES position. Set the STATION SELECTOR on Position 1 and depress the METER switch. Repeat this operation with the STATION SELECTOR at Positions 2, 3, and 4. Movement of the meter needle into the red area of the meter dial in any instance indicates insulation resistance less than 50 megohms. Apply a pressure of 261 kPa (37.8 PSIG) to the pressure pot and set the STATION SELECTOR on Position 6. Depress the METER switch. Movement of the meter needle into the red area indicates insulation resistance less than 50 megohms. Repeat this test with the STATION SELECTOR on Positions 7 through 20.

4.7.11.2.2 Mark 41 Mod 1 (M114). Place each assembled switch within a conditioning chamber chamber for a period of 24 hours while maintaining the chamber at a temperature of 25° ± 1°C (77°F ± 2°F) and a relative humidity of 75 ± 5 percent. Remove the switch from the chamber and place the switch in the pressure pot of the Test Set Mark 250 in the same manner as in 4.7.10. The pressure in the pot shall remain at atmospheric during the first part of this test. Plug the Cable Assembly CA-963 into the receptacle of the switch and rotate the TEST SELECTOR to the INSULATION RES position. Set the STATION SELECTOR on Position 1 and depress the METER switch. Repeat this operation with the STATION SELECTOR at Positions 2, 3, and 4. Movement of the meter needle into the red area of the meter dial in any instance indicates insulation resistance less than 50 megohms. Apply a pressure of 62 kPa (9.0 PSIG) to the pressure pot and set the STATION SELECTOR on Position 6. Depress the METER switch. Movement of the meter needle in the red area indicates insulation resistance less than 50 megohms. Repeat this test with the STATION SELECTOR on Positions 7 through 20.

4.7.11.3 Contact resistance test.

4.7.11.3.1 Mark 41 Mod 0 (M115). With the hydrostatic switch mounted in the Test Set Mark 250, rotate the TEST SELECTOR to the CONTACT RES position. Increase the pressure in the pressure pot to 261 kPa (37.8 PSIG). At this pressure make the following test: Rotate the STATION SELECTOR to Position 1 and depress the METER switch. Movement of the meter needle into the red area of the meter dial indicates contact resistance in excess of 0.2 ohm. Repeat the test with the STATION SELECTOR at Positions 2, 3, and 4. Movement of the meter needle into the red area of the meter dial in any instance indicates contact resistance in excess of 0.2 ohm. Upon completion of the foregoing test, reduce the pressure in the pressure pot to atmospheric and remove the switch from the pressure pot.

4.7.11.3.2 Mark 41 Mod 1 (M116). With the hydrostatic switch mounted in the Test Set Mark 250, rotate the TEST SELECTOR to the CONTACT RES position. Increase the pressure in the pressure pot to 62 kPa (9.0 PSIG). At this pressure make the following test: Rotate the STATION SELECTOR to position 1 and depress the METER switch. Movement of the meter needle into the red area of the meter dial indicates contact resistance in excess of 0.2 ohm. Repeat the test with the STATION SELECTOR at Positions 2, 3, and 4. Movement of the meter needle into the red area of the meter dial in any instance indicates contact resistance in excess of 0.2 ohm. Upon completion of the foregoing test, reduce the pressure in the pressure pot to atmospheric and remove the switch from the pressure pot.

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4.8 Environmental test. Divide the sample into three equal groups. See Table I for test sequences.

4.8.1 Vibration. Securely fasten the switch directly to the vibration table by either simulating the actual method of mounting used in service, or by its external case, whichever is appropriate. Apply simple harmonic excitation parallel to each of the three principal axes of the switch for equal periods of time. The frequency range from 10 to 60 Hz shall be covered either by slowly cycling at a logarithmic rate between the limits, or by at least 24 discrete frequency steps which have a logarithmic distribution. Maintain the vibration amplitude at 2 ± .2g or 2.5 ± 0.3mm (0.10 ± .01 inch) (peak-to-peak), whichever is the lesser. The duration of the test in each direction is four hours. The total test duration shall be twelve hours.

4.8.2 Shock. Subject the switch to a continuous two-phase shock applied in any direction with respect to the hydrostatic switch as follows:

a. The first (impact) phase shall consist of a change of velocity of 1.7 to 2.1 m/s (5.5 to 7.0 feet per second) taking place in from 0.2 to 0.4 millisecond.

b. The second (drag) phase shall consist of accelerating the switch for a minimum time of 15 milliseconds in such a manner that during this time the minimum average acceleration is 90 gravity units with the peak acceleration between 125 and 150 gravity units.

4.8.3 Temperature.

4.8.3.1 Operating. Subject the switches to the tests of 4.7.10, 4.7.11.1, 4.7.11.2, and 4.7.11.3 at temperatures of -2° ± 1°C (28° ± 2°F) and 32° ± 2°C (90° ± 3°F).

4.8.3.2 Storage. Subject the switches, packaged and packed as designated in Section 5, to one 14-day JAN TEMPERATURE AND HUMIDITY CYCLE defined in MIL-STD-331, Test Method 105, except that the relative humidity in the chamber need not be raised above room ambient or ambient as controlled.

4.8.4 High pressure tests.

4.8.4.1 Ruggedness test (M117). With safety pin removed, mount the hydrostatic switch with the Test Chamber, Item 4.3.c., and apply to the sea side of the switch a 4.14 MPa (600 PSIG) hydrostatic pressure for no less than 3 minutes. Observe the switch for evidence of leakage. See Table I.

4.8.4.2 Safety test (M118). With the safety pin in place, mount the hydrostatic switch within the Chamber, Item 4.3c., and apply to the sea side of the hydrostatic switch a 4.14 MPa (600 PSIG) hydrostatic pressure for no less than 3 minutes. Verify that the contacts of the switch do not close during the test.

4.9 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Major or Minor in accordance with WR-43. Tests and examinations that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified Minor, AQL 2.5 percent.

5 PREPARATION FOR DELIVERY (See 6.2h)

5.1 Preservation and packaging. Each hydrostatic switch shall be individually packaged in accordance with DL 2644479, unless otherwise specified in the contract or order.

5.2 Packing. Twenty-four hydrostatic switches individually packaged as specified under 5.1 shall be packed in accordance with DL 2644484.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with MIL-STD-129.

6 NOTES

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6.1 Intended use. The hydrostatic switch is intended for use in underwater mines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification; and the "mod" of the switches being procured.

b. Number of first articles required. (3.2, 4.4.1)

c. That the standards of workmanship exhibited by the previously approved production sample shall be used to determine the minimum requirements of the current contract relative to workmanship, when first article sample requirements are waived because of having been submitted on a recent contract. (3.7)

d. Test equipment to be loaned by the Government, and the terms under which it is loaned. (4.3a)

e. Activity to perform first article testing, if other than contractor. (4.4)

f. Lot size for sampling purposes, if one month's scheduled production is not acceptable. (4.5.2)

g. Activity to perform periodic production testing, if other than contractor. (4.6)

h. That preparation for delivery shall be adequate to protect the switch prior to its installation, when switches are purchased for a prime contractor for installation into Assembly Condition "D" mines.

i. Number of units in the periodic production sample.

6.3 Definitions. The following terms are used throughout this specification and are here defined for clarity.

6.3.1 Hydrostatic switch. Reference herein to the switch or hydrostatic switch shall be understood to mean the Switch, Hydrostatic, Mark 41 Mod 0 or 1.

6.3.2 Lot. Unless otherwise specified in the contract or purchase order, lot size shall mean one month's scheduled production. See Section 5 of MIL-STD-105 for lot composition.

6.4 Metrication. This specification utilizes metric units in accordance with ASTM E 380-72 as approved by the Department of Defense. Quantities expressed in conventional units are noted parenthetically. Of particular interest are the metric units of force and pressure, the newton (N) and pascal(Pa), respectively. One pound (force) equal 4.45 newtons and one psi equals 6.895 pascals.

6.5 The design and technical cognizance for the switches specified herein are located at the Naval Mine Engineering Facility, Yorktown, Virginia 23691.

Custodian:
NAVY - OS

Preparing Activity:
NAVY - OS
Project Number
1350-N029

Review Activities
NAVY - OS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-21920C(OS), Switch, Hydrostatic, Mark 41 Mods 0 and 1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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